
PROJECTS

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RULES (1/2)

- Projects are individual
 - If your project requires code (i.e., it is a practical application) you only have to submit the code
 - If your project is a review of the literature or a theoretical result you only have to submit a report
 - During the oral exam you will have about 15 minutes to present your project. You can use slides, notebooks, etc.
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RULES (2/2)

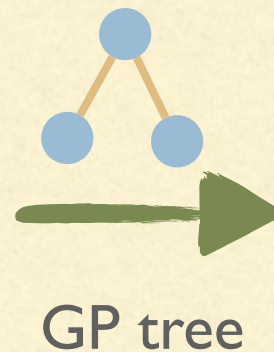
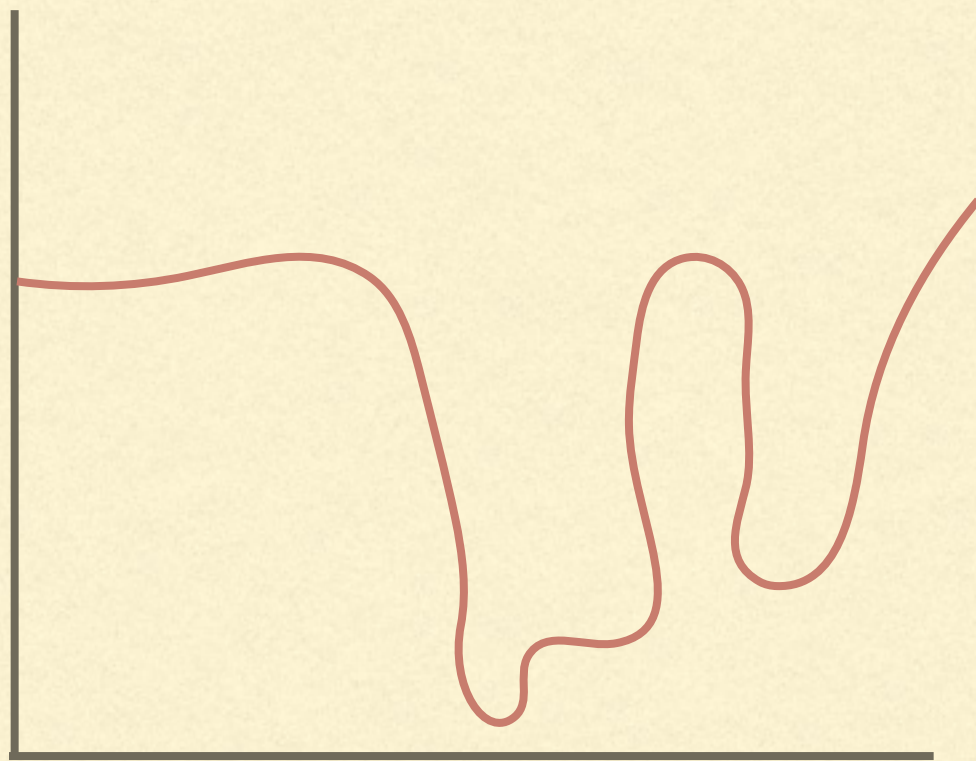
- Some of the projects require to create something new that may or may not work, so even negative results can be presented
 - Some of the projects can be extended to become a full thesis
 - You can ask for guidance before completing the project
 - Once you select a project write what project you selected on the team of the course
 - If you want you can change the project just ask
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PROBLEM TRANSFORMATION USING GP

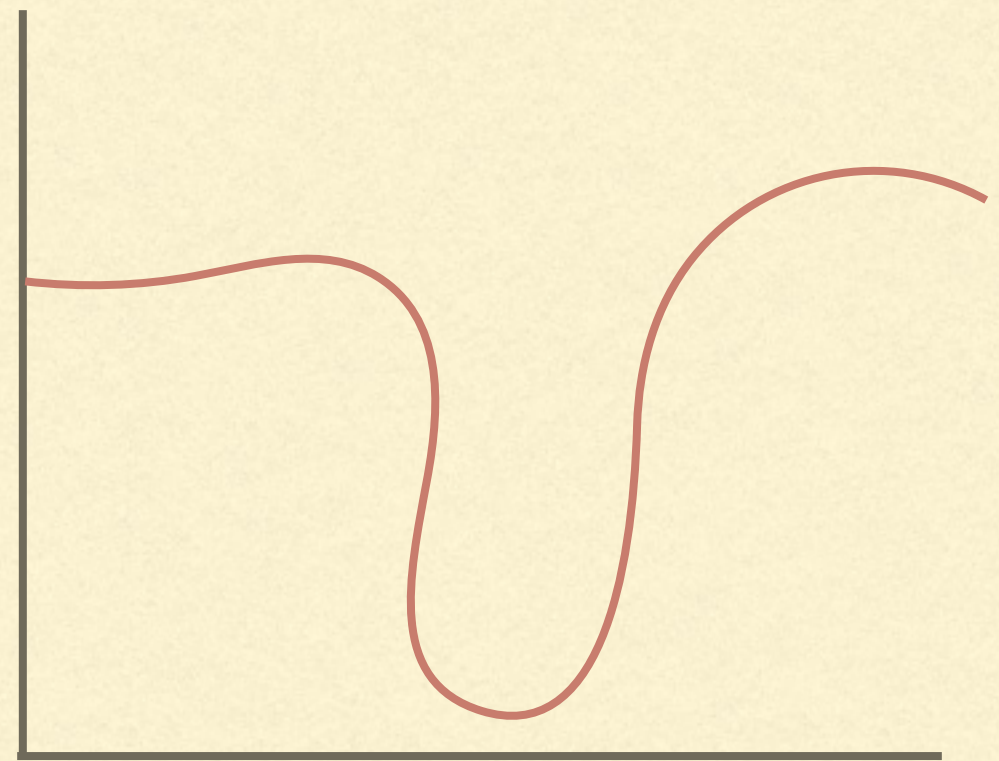
Evolve via GP a transformation of a problem that make it easier for another optimization algorithm, like $(\mu + \lambda) - ES$ or PSO.

Either the mapping should be invertible or the optimum should not change position “too much”

Original landscape



Remapped landscape



CELLULAR GENETIC PROGRAMMING

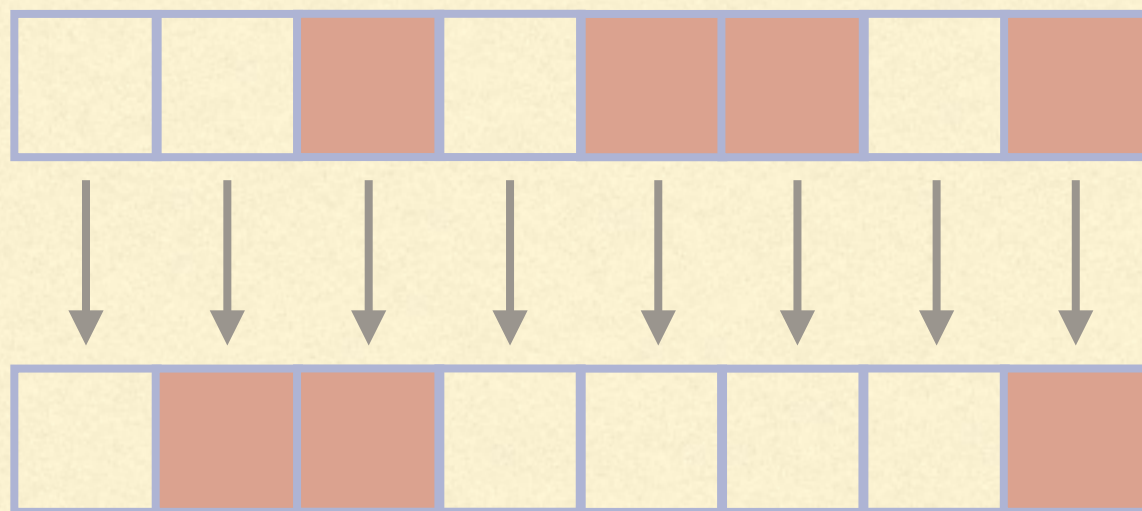
- Put individuals of a population on a grid (or any graph!)
 - Evolve the individual using only local information
 - What happens for different topologies?
 - Test at least grids (in multiple dimensions)
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CODE OPTIMIZATIONS WITH GE

- Start with a snippet working code in a language of your choice (start with a simple one or a subset of another language)
 - Use Grammatical Evolution to find alternative versions of the code using as fitness a combination of:
 - Is the code working?
 - How fast is the code?
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EVOLUTION OF CELLULAR AUTOMATA RULES WITH GP

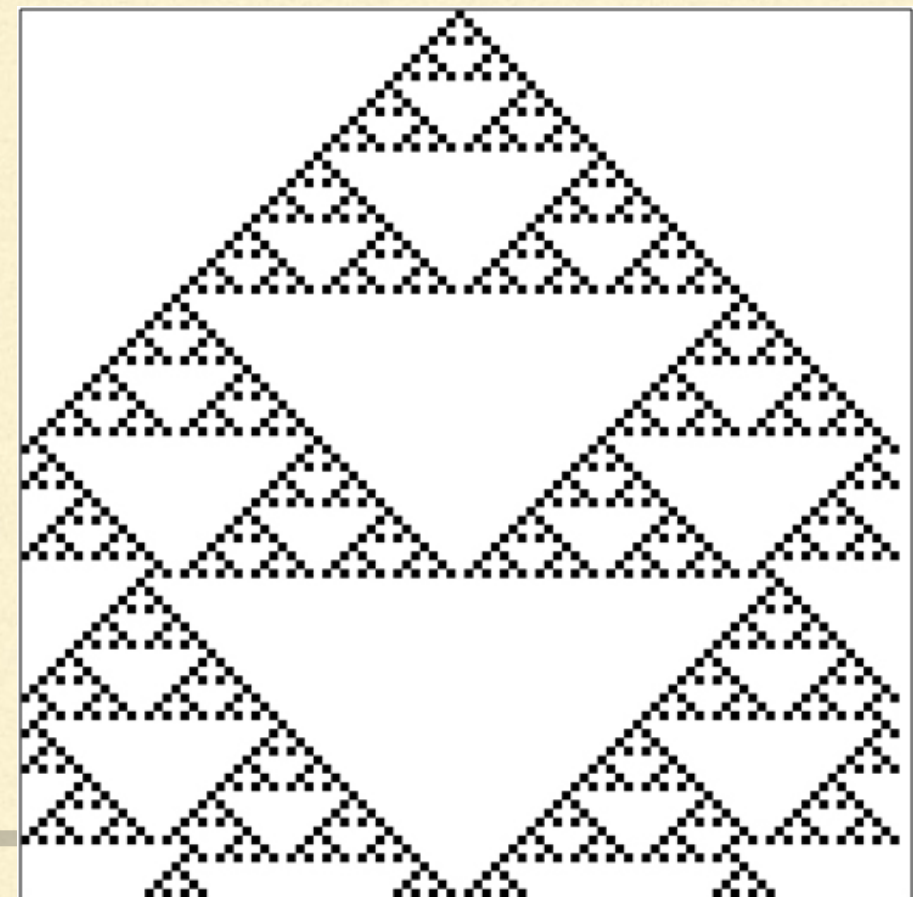
A Cellular Automata (CA) is a collection of cells with a state from a finite set (e.g. $A = \{0,1\}$)



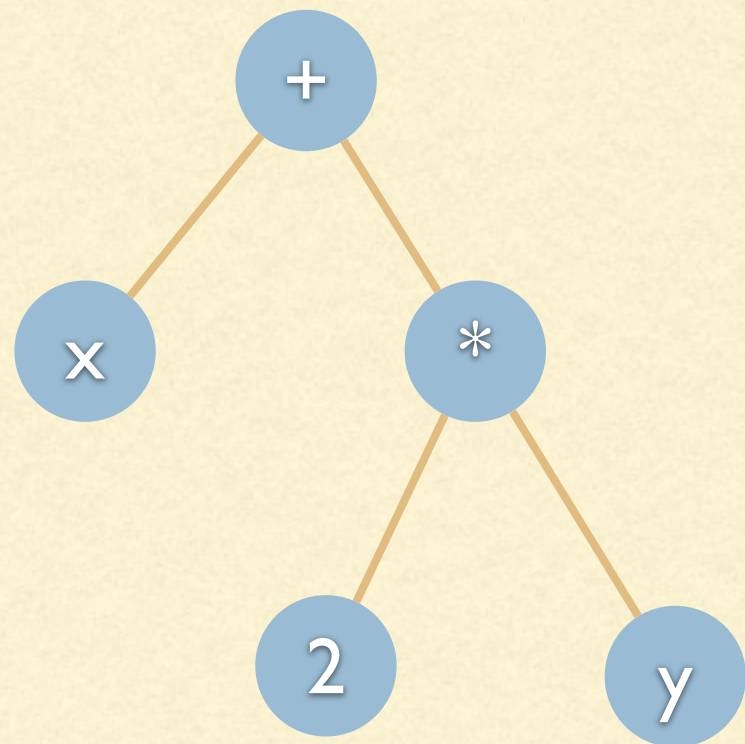
Very complex behaviors can be obtained by simple rules

Can we evolve the local rule of a CA to conform to a given behavior?

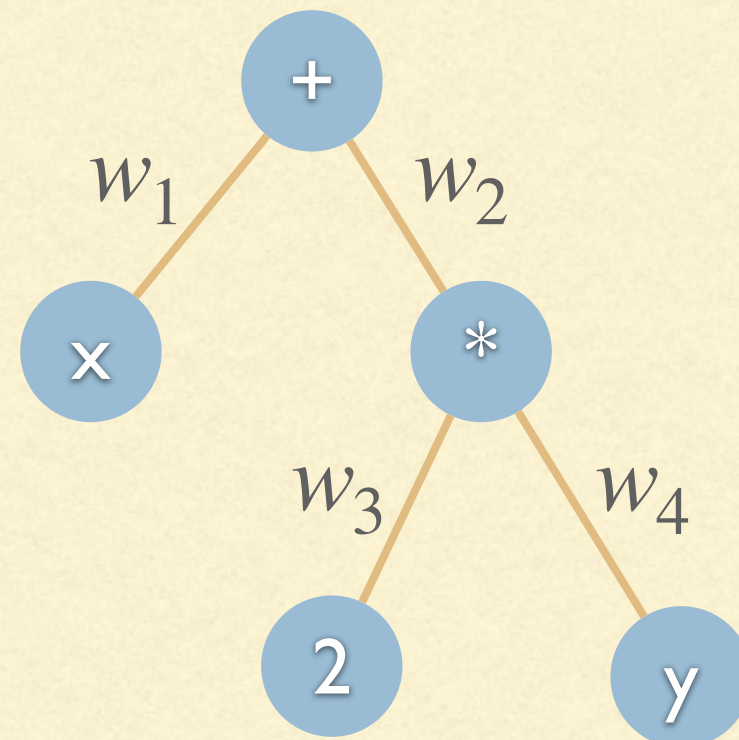
At each time step all the cells update their status by looking only on a local neighborhood using a so called “local rule”.



COMBINING GP AND GRADIENT DESCENT



$$x + 2y$$



$$w_1x + w_2(2w_3 + w_4y)$$

The parameter $w_1, \dots, w_4$ (that initially can be 1)
can now be optimized via gradient descent
(or any gradient-based method)

HYBRID PSO-GA

- GA works well on discrete spaces
 - PSO works well on continuous spaces
 - Can we devise a hybrid PSO-GA algorithm in which the continuous part “moves” like PSO and the discrete part evolves like GA?
 - Mutation should not be problematic, crossover might be a little bit more complex
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CO-EVOLUTION OF ROBOTS

- Like “robot wars”, two robots are fighting against each other and the winner is the surviving one
 - Define a set of actions (move, shoot, etc.) and sensors inputs
 - Co-evolve GP individuals that controls the robots
 - Also possible using small neural networks with NEAT
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EVO SUPER MARIO

- This code evolves a neural network to play Super Mario using NEAT: <https://github.com/vivek3141/super-mario-neat>
 - Can Cartesian GP or Linear GP be used? How are their performances?
 - If you have in mind other games and you want to create a game to play them you can do it!
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EVOLVING DIFFERENTIAL EQUATIONS

- Take a physical system governed by a set of differential equations
 - Generate a collection of data from the system (using a simulation)
 - Try to evolve the original equations with GP (notice: what variables do you need? What operators?)
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PSO/DE/ES FOR INVERSE PROBLEMS

- The main idea is that you have a dynamical system for which you know the equations describing it
 - The equations have some parameters that you do not know
 - You have data on the dynamical behavior of the system
 - Which parameters gave that dynamical behavior?
 - We can find them with some evolutionary or swarm intelligence method
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CO-EVOLUTION AND GAME THEORY

- Explore how different scenario that happens in game theory (like Nash Equilibria) appears in co-evolution scenario
 - Possibilities:
 - A theoretical work, exploring the link between game theory and evolutionary computation
 - A practical approach with simulations (and possibly links with the area of “artificial life”)
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